ANIL NEERUKONDA INSTITUTE OF TECHNOLOGY & SCIENCES

Autonomous Institution Affiliated to Andhra University | Approved by AICTE

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Course: Full Stack Web Development Lab

Lab Record Week: #03

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1. TITLE OF THE EXPERIMENT

JavaScript ES6 Array Methods and Object-Oriented Programming – Library Management

2. AIM

To implement modular JavaScript functions for array operations (calculating total, average, adding marks) and develop an Object-Oriented Library Management System using ES6 classes with methods for adding, displaying, totaling, and removing books.

3. PROCEDURE / ALGORITHM

Step 1: Create function.js defining an array of student marks.

Step 2: Implement displayMarks(), totalMarks(), averageMarks(), and addMark() functions.

Step 3: Test array manipulation and print updated values to console.

Step 4: Create object.js defining an ES6 Library class with constructor initializing an empty books array.

Step 5: Implement addBook(name, price), displayBooks(), totalPrice(), and removeBook() methods.

Step 6: Instantiate Library object, add multiple books, calculate total catalog value, remove a book, and print final output.

4. TEST CASES / VERIFICATION

Test Case	Description / Action	Expected Output	Status
TC-01	Array Calculation Functions	Compute correct total (436) and average (87.2) of marks array	**Passed**
TC-02	Array Element Insertion	Append new mark (92) and recalculate updated total (528)	**Passed**
TC-03	ES6 Library Class Methods	Add books (Python, Java, HTML, JS) and display catalog list	**Passed**
TC-04	Catalog Total & Removal	Compute total price (■1800) and remove last book correctly	**Passed**

5. LEARNING OUTCOME / RESULT

JavaScript array manipulation functions and the ES6 Library class were successfully implemented and verified with accurate terminal and console outputs.

6. SOURCE CODE

File: function.js

```
// Array
let marks = [85, 90, 78, 88, 95];

// Function to display marks
function displayMarks(arr) {
    console.log("Student Marks:", arr);
}

// Function to calculate total
function totalMarks(arr) {
    let total = 0;

    for (let i = 0; i < arr.length; i++) {
        total += arr[i];
    }

    return total;
}

// Function to calculate average
function averageMarks(arr) {
    return totalMarks(arr) / arr.length;
}

// Function to add a new mark
function addMark(arr, mark) {
    arr.push(mark);
    return arr;
}

displayMarks(marks);

console.log("Total Marks:", totalMarks(marks));

console.log("Average Marks:", averageMarks(marks));

console.log("After Adding New Mark:");

addMark(marks, 92);

displayMarks(marks);

console.log("New Total:", totalMarks(marks));
```

File: object.js

```
console.log("===== Library Management System =====");

class Library {

    constructor() {
        this.books = [];
    }

    // Add Book
    addBook(name, price) {
        this.books.push({
            title: name,
            cost: price
        });

        console.log(name + " book added successfully.");
    }

    // Display Books
    displayBooks() {
```

```
        console.log("\nBooks in Library:");

        for (let i = 0; i < this.books.length; i++) {

            console.log(
                (i + 1) + ". " +
                this.books[i].title +
                " - ■" +
                this.books[i].cost
            );

        }

    }

    // Calculate Total Price
    totalPrice() {

        let total = 0;

        for (let i = 0; i < this.books.length; i++) {

            total += this.books[i].cost;

        }

        return total;

    }

    // Remove Last Book
    removeBook() {

        let removedBook = this.books.pop();

        console.log("\nRemoved Book: " + removedBook.title);

    }

}

const lib = new Library();

lib.addBook("Python", 500);
lib.addBook("Java", 600);
lib.addBook("HTML", 300);
lib.addBook("JavaScript", 400);

lib.displayBooks();

console.log("\nTotal Price: ■" + lib.totalPrice());

lib.removeBook();

console.log("\nAfter Removing One Book:");

lib.displayBooks();

console.log("\nNew Total Price: ■" + lib.totalPrice());
```

7. OUTPUT SCREENSHOTS

Output Screenshot: Screenshot 2026-08-25 061620.png

```
PS C:\Users\manoj\OneDrive\Desktop\fs_lab\records\week3> node function.js
Student Marks: [ 85, 90, 78, 88, 95 ]
Total Marks: 436
Average Marks: 87.2
After Adding New Mark:
Student Marks: [ 85, 90, 78, 88, 95, 92 ]
New Total: 528
```

Output Screenshot: Screenshot 2026-08-25 061727.png

```
PS C:\Users\manoj\OneDrive\Desktop\fs_lab\records\week3> node object.js
===== Library Management System =====
Python book added successfully.
Java book added successfully.
HTML book added successfully.
JavaScript book added successfully.

Books in Library:
1. Python - ₹500
2. Java - ₹600
3. HTML - ₹300
4. JavaScript - ₹400

Total Price: ₹1800

Removed Book: JavaScript

After Removing One Book:

Books in Library:
1. Python - ₹500
2. Java - ₹600
3. HTML - ₹300

New Total Price: ₹1400
```